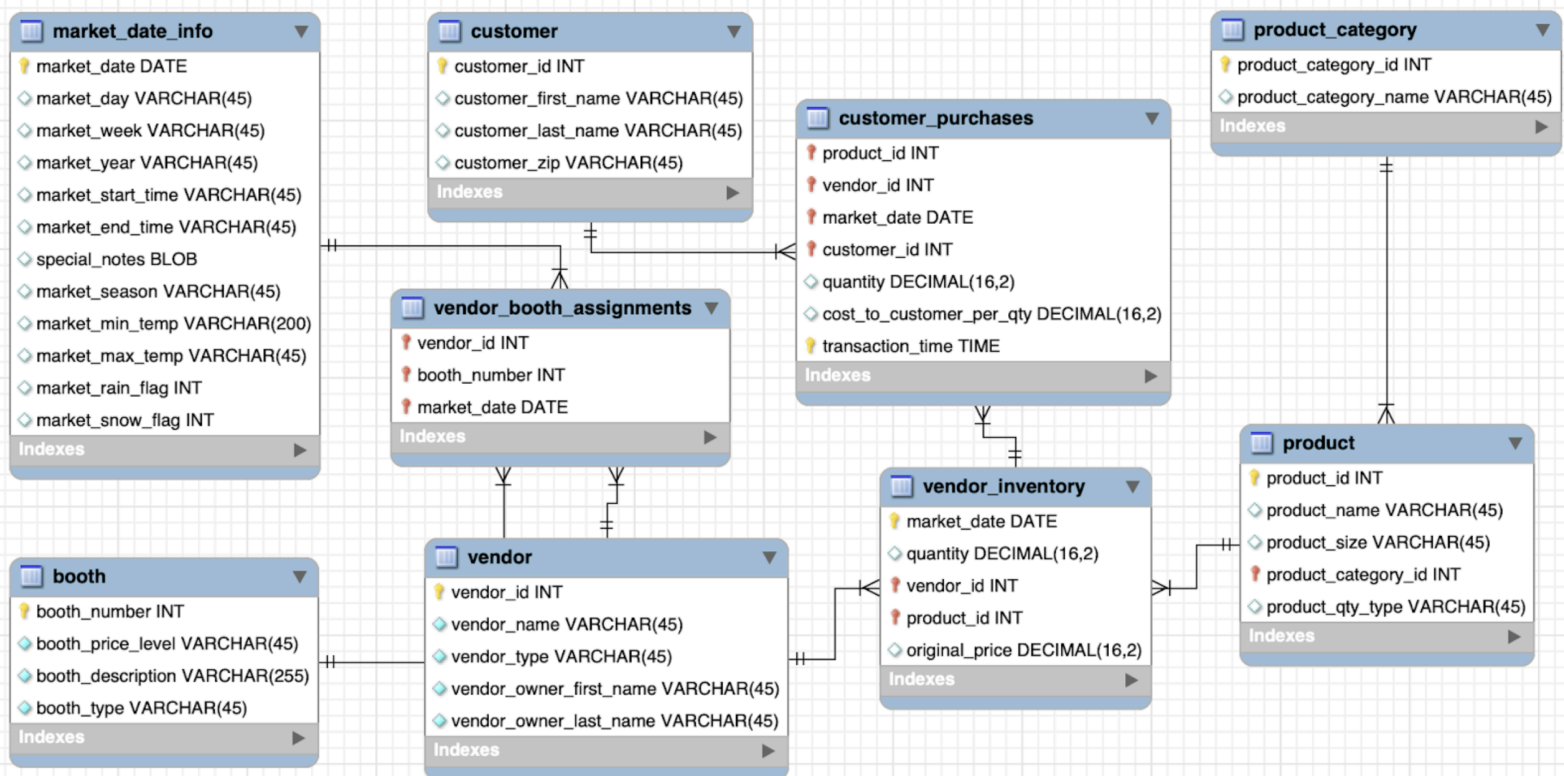




WhatsApp link

<https://chat.whatsapp.com/EvVhJ555AGW1cJaeKI0USe>

Dataset



CASE Statements

Syntax :

Case

When condition then value

When condition then value

else value

End as alias

Questions on CASE Statement

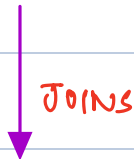
Question: Find out which vendors primarily sell fresh produce and which don't.

Question: Put the total cost for each transaction in customer purchases into bins of

- under \$5.00
- 5.00–9.99,
- 10.00–19.99
- \$20.00 and over.

Why Joins when you have subquery?

- ① Subquery will be bound to show the columns from outer query.
- ② Subqueries are generally^{*} less performant
↳ there are some exceptions
- ③ Subqueries with multiple levels are always difficult to manage and read



- ① You can showcase columns from multiple tables using Joins
- ② Joins while performed on indexes are almost always^{*} performant
- ③ Easier to maintain

Join Visualizer

<https://docs.google.com/spreadsheets/d/1xNLbRskqZL8L2OyjdcGIU3rfn2Lzhn9nie1NBrubTsl/edit?usp=sharing>

Syntax for Joins

PK					FK			
	eid	name	salary	dept_id	dept_id	name	location	
	1	A	12L	101	101	IT	India	
	2	B	7L	102	102	Sales	India	
	3	C	100L	103	103	Leadership	USA	
	4	D	5L	null				

eid	e name	salary	dept_id	d name	location
1	A	12L	101	IT	India
2	B	7L	102	Sales	India
3	C	100L	103	leader	USA

① Joins are horizontal merges

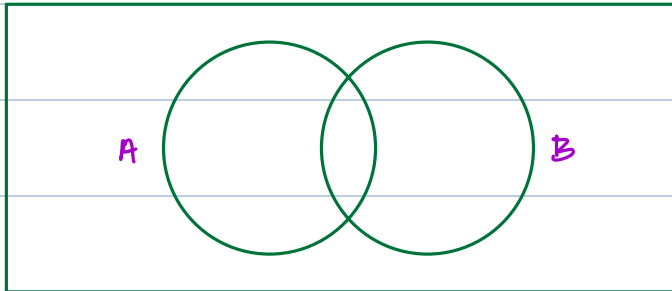
② normally Joins are dependent on a single column value in both the tables

we will have to use alias

Select eid, ename, salary, dept_id, dname, location
from employees Join Department
on employees.dept_id = Department.dept_id

Types of Join

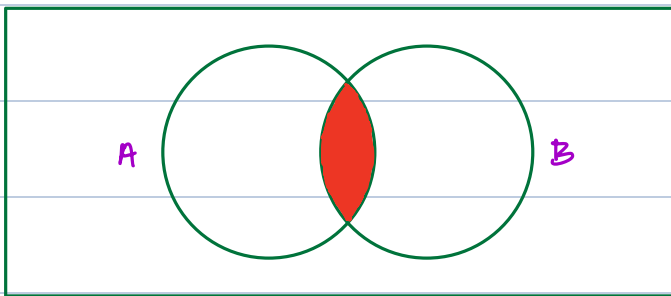
For visualization we will be using Venn diagrams



$A = \{1, 2, 3, 4, 5\}$

$B = \{4, 5, 6, 7, 8\}$

① Inner Join / Default Join / Natural Join



only the common values
will be selected

$A = \{1, 2, 3, 4, 5\}$ $\rightarrow \{4, 5\}$

$B = \{4, 5, 6, 7, 8\}$

Syntax

Select *

from A

inner join B

on $A.id = B.id$

example 1

A

B

w	x		y	z		w	x	y	z
1	a		3	m		3	c	3	m
2	b		4	n		4	d	4	n
3	c		5	o					
4	d		6	p					

on $A.w = B.y$

example 2

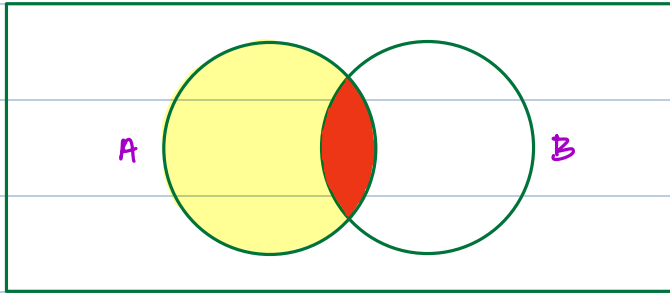
A

B

w	x		y	z		w	x	y	z
1	a		2	m		2	b	2	m
2	b		3	n		2	c	2	m
2	c		3	o		3	d	3	n
3	d		4	p		3	d	3	o
4	e		null	q		4	e	4	p
null	f								

on $A.w = B.y$

② Left Join / Left Outer Join



$A = \{1, 2, 3, 4, 5\}$

$B = \{4, 5, 6, 7, 8\}$

A left join B

$\{1, 2, 3, 4, 5\}$

Syntax

Select *

from A

left join B

on A.id = B.id

Note: When to use Left Join vs when to use Inner join

all, every, each → left
else → inner join

99%

example 1

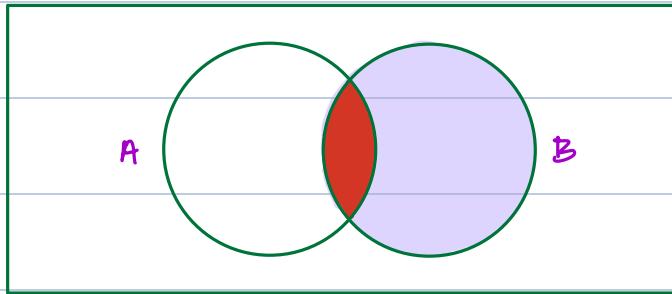
A		B								
w	x		y	z			w	x	y	z
1	a		3	m			1	a	null	null
2	b		4	n			2	b	null	null
3	c		5	o		→	3	c	3	m
4	d		6	p			4	d	4	n

$$A.w = B.y$$

example 2

w	x		y	z		w	x	y	z
1	a		2	m		1	a	null	null
2	b		3	n	→	2	b	2	m
2	c		3	o		2	c	2	m
3	d		4	p		3	d	3	n
4	e		null	q		3	d	3	o
null	f					4	e	4	p
						null	f	null	

Right Join / Right Outer Join



No right join

In my production

$A = \{1, 2, 3, 4, 5\}$

$B = \{4, 5, 6, 7, 8\}$

Syntax

Select *

from A

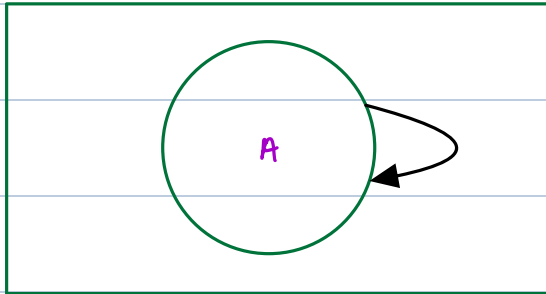
right join B

on A.id = B.id

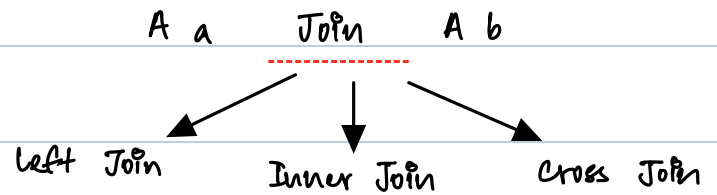
Note

- ① You can convert any right join as left join by changing the table sides

Self Join



Join with the same table



$$1 \cdot 2 = 2 \cdot 1$$

left table, size id = right table.p_id

80% of our code

→ Get Manager Name

Cross Join / Cartesian Product

$A = \{1, 2, 3\}$

$B = \{4, 5\}$

$A \times B =$

$\{1, 4\}$ $\{1, 5\}$

$\{2, 4\}$ $\{2, 5\}$

$\{3, 4\}$ $\{3, 5\}$

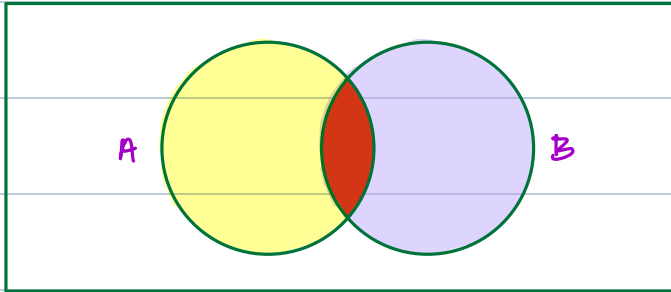
	Car_model
1	Camry
2	Corolla
3	Prius

	Color_name
1	Black
2	Red
3	Silver

CROSS JOIN

	Car_model	Color_name
1	Camry	Black
2	Corolla	Black
3	Prius	Black
4	Camry	Red
5	Corolla	Red
6	Prius	Red
7	Camry	Silver
8	Corolla	Silver
9	Prius	Silver

Full Join / Full Outer Join



No direct way of doing
this in MySQL

Left Join Union Right Join

w	x		y	z		w	x	y	z
1	a		3	m					
2	b		4	n					
3	c		5	o					
4	d		6	p					